

1) Uporabimo izrazu dostanemo:

$$\frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2) = \frac{1}{4} \left(\left(\sqrt{\langle \vec{u} + \vec{v}; \vec{u} + \vec{v} \rangle} \right)^2 - \left(\sqrt{\langle \vec{u} - \vec{v}; \vec{u} - \vec{v} \rangle} \right)^2 \right) \stackrel{\substack{\text{linearnost v 2. zlo\u017ekke; skorsimetrija}}}{=} \\ = \frac{1}{4} (\langle \vec{u}; \vec{u} \rangle + \langle \vec{u}; \vec{v} \rangle + \langle \vec{v}; \vec{u} \rangle + \langle \vec{v}; \vec{v} \rangle - \langle \vec{u}; \vec{u} \rangle + \langle \vec{u}; \vec{v} \rangle + \langle \vec{v}; \vec{u} \rangle - \langle \vec{v}; \vec{v} \rangle) = \\ = \frac{1}{4} (2\langle \vec{u}; \vec{v} \rangle + 2\langle \vec{v}; \vec{u} \rangle) \stackrel{\substack{\text{skorsimetrija}}}{=} \frac{1}{4} \cdot 2 (\langle \vec{u}; \vec{v} \rangle + \overline{\langle \vec{u}; \vec{v} \rangle}) = \frac{1}{2} (2 \operatorname{Re} (\langle \vec{u}; \vec{v} \rangle)) = \operatorname{Re} (\langle \vec{u}; \vec{v} \rangle)$$

$$\frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2) = \frac{1}{4} (\langle \vec{u} - i\vec{v}; \vec{u} - i\vec{v} \rangle - \langle \vec{u} + i\vec{v}; \vec{u} + i\vec{v} \rangle) = \\ = \frac{1}{4} (\langle \vec{u}; \vec{u} \rangle - \langle \vec{u}; i\vec{v} \rangle - \langle i\vec{v}; \vec{u} \rangle + \langle i\vec{v}; i\vec{v} \rangle - \langle \vec{u}; \vec{u} \rangle - \langle \vec{u}; i\vec{v} \rangle - \langle i\vec{v}; \vec{u} \rangle - \langle i\vec{v}; i\vec{v} \rangle) = \\ = \frac{1}{4} (-2(\langle \vec{u}; i\vec{v} \rangle + \langle i\vec{v}; \vec{u} \rangle)) = -\frac{1}{2} (i\langle \vec{u}; \vec{v} \rangle + \overline{i\langle \vec{u}; \vec{v} \rangle}) = -\frac{1}{2} (i\langle \vec{u}; \vec{v} \rangle - i\overline{\langle \vec{u}; \vec{v} \rangle}) = \\ = -\frac{1}{2} i (\langle \vec{u}; \vec{v} \rangle - \overline{\langle \vec{u}; \vec{v} \rangle}) = -\frac{1}{2} i \cdot 2i \operatorname{Im} (\langle \vec{u}; \vec{v} \rangle) = \operatorname{Im} (\langle \vec{u}; \vec{v} \rangle)$$

QED